model to induce robustness overfitting, the model fine-tuned with downstream data tends to be less robust, especially compared to a downstream model directly enhanced with AT. Therefore, we opt for $\mathcal{I}_B$ as our inference strategy. A comprehensive comparison between $\mathcal{I}_A$ and $\mathcal{I}_B$ can be found in Section C.

5. Evaluation

5.1. Experimental Setup

Datasets. We use the following five datasets for evaluation.
• *CIFAR-100* [30]: This dataset contains $50,000$ training images and $10,000$ testing images. Each image has a size of $32 \times 32 \times 3$ and belongs to one of 100 classes.
• *Tiny ImageNet* [33]: This dataset contains $100,000$ training images and $10,000$ testing images. Each image has a size of $64 \times 64 \times 3$ and belongs to one of 200 classes.
• *CelebA* [38]: This dataset contains $202,599$ face images annotated by 40 attributes. In our experiment, we train a eight-class classifier to predict the top three most balanced attributes including Heavy Makeup, Mouth Slightly Open, and Smiling.
• *STL-10* [11]: This dataset contains $13,000$ color images. Each image has a size of $96 \times 96 \times 3$ and belongs to one of 10 classes, and each class contains $1,300$ images.
• *CINIC-10* [12]: This dataset is compiled from images selected and downsampled from the ImageNet database [14]. This dataset includes $270,000$ color images.

CIFAR-100 and Tiny ImageNet serve as the upstream datasets, while CelebA, STL-10, and CINIC-10 are used as the downstream datasets. Each original downstream dataset, denoted by $\mathcal{D}$, is partitioned into four equally sized, disjoint subsets: $\mathcal{D}^{\text{train}}_{\text{target}}$, $\mathcal{D}^{\text{test}}_{\text{target}}$, $\mathcal{D}^{\text{train}}_{\text{shadow}}$, and $\mathcal{D}^{\text{test}}_{\text{shadow}}$. The subsets $\mathcal{D}^{\text{train}}_{\text{target}}$ and $\mathcal{D}^{\text{test}}_{\text{target}}$ serve as the training and testing sets for the target model, respectively, while $\mathcal{D}^{\text{train}}_{\text{shadow}}$ and $\mathcal{D}^{\text{test}}_{\text{shadow}}$ are used as the training and testing sets for the shadow model.

Pre-training procedure. We use CIFAR-100 and Tiny ImageNet as our pre-training datasets and employ ARO to train a PreAct-ResNet18 [19] model as the pre-trained model. We compute the mean and standard deviation on the training set to normalize the input images. We adopt the cross-entropy loss function and the SGD optimizer with momentum with a step-wise learning rate decay. Specifically, the model is trained for 200 epochs with the learning rate initialized at 0.01 and divided by 10 after 100 and 150 epochs. The batch size is set to 128, and the weight decay is set to 1×10^{-4}.

Fine-tuning procedure. Given a pre-trained model, we use it to train classifiers for the downstream datasets – CelebA, STL-10, and CINIC-10. We utilize the pre-trained model's parameters as the initial parameters of the downstream classifier and then train the classifier on the downstream dataset. We adopt the cross-entropy loss function and the SGD optimizer with momentum when training a downstream classifier. The model is fine-tuned for 100 epochs with the learning rate initialized at 0.01 and divided by 10 after 50 and 75 epochs. The momentum is set to 0.9, and the weight decay is set to 5×10^{-4} for all experiments.

MIA configuration and evaluation metrics. According to the different knowledge levels the attacker has about the downstream model, we consider four types of MIAs:
• *Black-Box/Shadow* [50]: In this scenario, the attacker only has black-box access to the target model with a local shadow dataset and lacks knowledge of the training dataset of the target model. Note that the attacker can obtain confidence scores returned by the target model.
• *Black-Box/Partial* [50]: The attacker has black-box access to the target model and a partial training dataset of the target model. In this scenario, the adversary does not need to train a shadow model; rather, they use the partial training dataset as the ground truth for membership and directly train their attack model.
• *White-Box/Shadow* [43]: The attacker has white-box access to the target model but only has a local shadow dataset and lacks knowledge of the training dataset of the target model. As the adversary has white-box access to the target model, they can also exploit the target sample's gradients with respect to the model parameters, embeddings from different intermediate layers, classification loss, and prediction posteriors.
• *White-Box/Partial* [43]: The attacker has white-box access to the target model and has a partial training dataset of the target model. The attack methodology here is almost identical to the black-box counterpart.

We then establish two types of attack models: one for the black-box and the other for the white-box setting. For black-box, our attack model has two inputs: the target sample's ranked posteriors and a binary indicator on whether the target sample being predicted correctly. Each input is first fed into a different 2-layer MLP (Multilayer Perceptron), then the two obtained embeddings are concatenated together and fed into a 4-layer MLP. For white-box, we have four inputs for this attack model, like the one used by Nasr et al. [43], including the target sample's ranked posteriors, classification loss, gradients of the parameters of the target model's last layer, and one-hot encoding of its true label. Each input is fed into a different neural network, and the resulting embeddings are concatenated together as input to a 4-layer MLP. We use ReLU as the activation function for the attack models. The batch size is set to 64, and cross-entropy is the loss function. We use Adam as the optimizer, with learning rate of 1e-5. The attack model is trained for 50 epochs. We adopt downstream accuracy and MIA's AUC (area under the ROC curve) score as the evaluation metrics.

Attack variants. We consider two variants of the proposed ARO attack to evaluate its effectiveness across different levels of aggression and stealthiness: a more stealthy version, referred to as ARO_α, and a more aggressive version, referred to as ARO_β. For ARO_α, we take L_2 threat model for adversarial training [40] with $\epsilon = 128/255$, step size $\alpha = 15/255$, and PGD steps $K = 1$. For overfitting perturbation magnitude, we adopt $\gamma = 1 \times 10^{-3}$ by default. Conversely, ARO_β employs the same parameters as ARO_α,